

Cognitive function in childhood, adolescence and early adulthood

Cognitive function was first tested at age 8 years in a school setting, and again in this setting at ages 11 and 15 years. A test of verbal comprehension given at the latter age was then re-administered at age 26 years (with 10 additional items of increased difficulty to avoid a ceiling effect), as part of the first interview of the survey member as an adult.

All coding to create the variables are described below and can be found in \$mr/cogchad/crsv.j.

Variables are located in the save file \$mr/cogchad/d081114.sav (which also includes SERNO, SEX and INF), and all cognitive variables are continuous.

Age 8 years

Raw variables (N.B. all these are included in the 'cogchad' save file)

VOC8R – a 50-item vocabulary test

SC8R – a 35-item verbal comprehension (sentence completion) test

R8R – mechanical reading (pronunciation) using the same words as for vocabulary

PI8R – a 60-item non-verbal ability (picture intelligence) test

For all 'R' variables 99 indicates missing data

VOC8N – as above for vocabulary but normalised to a mean of 100 and an SD of 15

SC8N - as above for sentence completion but normalised to a mean of 100 and an SD of 15

R8N - as above for mechanical reading but normalised to a mean of 100 and an SD of 15

PI8N - as above for picture intelligence but normalised to a mean of 100 and an SD of 15

For all 'N' variables 999 indicates missing data

Derived summary variable

COG8H – RH's standardised cognitive score at age8 - standardised for whole population available on each test see note below.

Formula

$$\text{COG8H} = (\text{VOC8H} + \text{SC8H} + \text{R8H} + \text{PI8H}) / 4$$

Where VOC8H, SC8H, R8H, PI8H are the standardised to the whole population available on each test versions of the variables VOC8R, SC8R, R8R, PI8R.

Descriptive statistics

	N	Min	Max	Mean	SD
SC8R	4259	0	34	14.20	7.78
R8R	4259	0	49	17.02	10.27
VOC8R	4259	0	40	16.28	5.99
PI8R	4266	0	60	40.20	9.48

13/05/10: Updated by SB to include COG8H

Descriptive Statistics

	N	Minimum	Maximum	Sum	Mean
cog8h cog score at 8 standardised(to whole popn on each test)	4256	-2.61	2.68	-2.78	-.0007
Valid N (listwise)	4256				

Age 11 years

Raw variables (N.B. all these are included in the 'cogchad' save file)

GA11R – an 80-item verbal and non-verbal ability test

V11R – the 40 verbal items from above

NV11R – the 40 non-verbal items from above

VOC11R – the same 50 words used at age 8 years for vocabulary

R11R - the same 50 words used at age 8 years for mechanical reading

A11R – a 50-item arithmetic test

For all 'R' variables 99 indicates missing data except GA11R, where 999 is used

GA11N – as above for the verbal and non-verbal ability test but normalised to a mean of 100 and an SD of 15

V11N - as above for the verbal ability test items but normalised to a mean of 100 and an SD of 15

NV11N - as above for the non-verbal ability test items but normalised to a mean of 100 and an SD of 15

VOC11N - as above for the vocabulary items but normalised to a mean of 100 and an SD of 15

R11N - as above for the mechanical reading items but normalised to a mean of 100 and an SD of 15

A11N - as above for the arithmetic items but normalised to a mean of 100 and an SD of 15

For all ‘N’ variables 999 indicates missing data

Derived summary variable

COG11H – RH’s standardised cognitive score at age11 - standardised for whole population available on each test see note below.

Formula

$$\text{COG11H} = (\text{R11H} + \text{VOC11H} + \text{GA11H} + \text{A11H}) / 4$$

Where R11H, VOC11H, GA11H, A11H are the standardised to the whole population available on each test versions of the variables R11R, VOC11R, GA11R, A11R

Descriptive statistics

	N	Min	Max	Mean	SD
GA11R	4032	0	80	45.01	15.88
V11R	4032	0	40	23.40	9.17
NV11R	4032	0	40	21.61	7.50
VOC11R	4027	0	49	29.99	7.45
R11R	4027	0	50	36.43	10.50
A11R	4025	0	50	26.39	11.74

13/05/10: Updated by SB to include COG11H

Descriptive Statistics

	N	Minimum	Maximum	Sum	Mean
cog11h cog score at 11 standardised(to whole popn on each test)	4024	-3.01	1.96	-.11	.0000
Valid N (listwise)	4024				

Age 15 years

Raw variables (N.B. all these are included in the 'cogchad' save file)

GA15R – the Heim AH4 Group Ability Test, a 130-item verbal and non-verbal ability test

V15R – the 65 verbal items from above

NV15R – the 65 non-verbal items from above

R15R – the Watts Vernon Reading Test, a 35-item verbal comprehension (sentence completion) test

M15R – a 47-item mathematics test

For all 'R' variables 99 indicates missing data, except for GA15R, where 999 is used

GA15N – as above for the Heim AH4 Group Ability Test but normalised to a mean of 100 and an SD of 15

V15N - as above for the verbal items of the AH4 but normalised to a mean of 100 and an SD of 15

NV15N - as above for the non-verbal items of the AH4 but normalised to a mean of 100 and an SD of 15

R15N - as above for the Watts Vernon but normalised to a mean of 100 and an SD of 15

M15N – as above for mathematics but normalised to a mean of 100 and an SD of 15

For all 'N' variables 999 indicates missing data

Derived summary variable

COG15H – RH's standardised cognitive score at age15 - standardised for whole population available on each test see note below.

Formula

$$\text{COG15H} = (\text{R15H} + \text{GA15H} + \text{M15H}) / 3$$

Where R15H, GA15H, M15H are the standardised to the whole population available on each test versions of the variables R15R, GA15R, M15R.

Descriptive statistics

	N	Min	Max	Mean	SD
GA15R	3982	0	125	73.84	20.04
V15R	4018	0	65	35.46	11.45
NV15R	4019	0	65	38.60	10.50
R15R	4014	0	35	24.47	6.6
M15R	4015	0	46	14.26	10.21

13/05/10: Updated by SB to include COG15H

Descriptive Statistics

	N	Minimum	Maximum	Sum	Mean
cog15h cog score at 15 standardised(to whole popn on each test)	4008	-2.93	2.17	-44.88	-.0112
Valid N (listwise)	4008				

Age 26 years

Raw variable (N.B. included in the 'cogchad' save file)

WV26R – the first 35 items of the Watts Vernon Reading Test, as used at age 15 years (missing = 99)

R26R – the Watts Vernon Reading Test used at 15 years, with an additional 10 items of increased difficulty to avoid a ceiling effect (missing = 99)

R26N – as above but normalised to a mean of 100 and an SD of 15 (missing = 999)

Descriptive statistics

	N	Min	Max	Mean	SD
WV26R	3715	0	35	28.76	5.4
R26R	3714	0	45	32.79	8.26